

=== creating venv ===

Requirement already satisfied: pip in ./venv/lib/python3.11/site-packages (22.3.1)

Collecting pip

Downloading pip-26.2.1-py3-none-any.whl (1.8 MB)

1.8/1.8 MB 91.8 MB/s eta 0:00:00

Installing collected packages: pip

Attempting uninstall: pip

Found existing installation: pip 22.3.1

Uninstalling pip-22.3.1:

Successfully uninstalled pip-22.3.1

Successfully installed pip-26.2.1

Looking in indexes: <https://download.pytorch.org/whl/cu121>

Collecting torch==2.5.1

Downloading torch-2.5.1%2Bcu121-cp311-cp311-linux_x86_64.whl (780.5 MB)

780.5/780.5 MB 92.2 MB/s 0:00:05

Collecting torchvision==0.20.1

Downloading torchvision-0.20.1%2Bcu121-cp311-cp311-linux_x86_64.whl (7.3 MB)

7.3/7.3 MB 156.4 MB/s 0:00:00

Collecting filelock (from torch==2.5.1)

Downloading filelock-3.32.3-py3-none-any.whl.metadata (2.0 kB)

Collecting typing-extensions>=4.8.0 (from torch==2.5.1)

Downloading typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)

Collecting networkx (from torch==2.5.1)

Downloading networkx-3.6.1-py3-none-any.whl.metadata (6.8 kB)

Collecting jinja2 (from torch==2.5.1)

Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)

Collecting fsspec (from torch==2.5.1)

Downloading fsspec-2026.7.0-py3-none-any.whl.metadata (10 kB)

Collecting nvidia-cuda-nvrtc-cu12==12.1.105 (from torch==2.5.1)

Downloading nvidia_cuda_nvrtc_cu12-12.1.105-py3-none-manylinux1_x86_64.whl (23.7 MB)

23.7/23.7 MB 512.8 MB/s 0:00:00

Collecting nvidia-cuda-runtime-cu12==12.1.105 (from torch==2.5.1)

Downloading nvidia_cuda_runtime_cu12-12.1.105-py3-none-manylinux1_x86_64.whl (823 kB)

823.6/823.6 kB 436.1 MB/s 0:00:00

Collecting nvidia-cuda-cupti-cu12==12.1.105 (from torch==2.5.1)

Downloading nvidia_cuda_cupti_cu12-12.1.105-py3-none-manylinux1_x86_64.whl (14.1 MB)

14.1/14.1 MB 598.6 MB/s 0:00:00

Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch==2.5.1)

Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl (664.8 MB)

664.8/664.8 MB 139.1 MB/s 0:00:02

Collecting nvidia-cublas-cu12==12.1.3.1 (from torch==2.5.1)

Downloading nvidia_cublas_cu12-12.1.3.1-py3-none-manylinux1_x86_64.whl (410.6 MB)

410.6/410.6 MB 571.2 MB/s 0:00:00

Collecting nvidia-cufft-cu12==11.0.2.54 (from torch==2.5.1)

Downloading nvidia_cufft_cu12-11.0.2.54-py3-none-manylinux1_x86_64.whl (121.6 MB)

121.6/121.6 MB 564.7 MB/s 0:00:00

Collecting nvidia-curand-cu12==10.3.2.106 (from torch==2.5.1)

Downloading nvidia_curand_cu12-10.3.2.106-py3-none-manylinux1_x86_64.whl (56.5 MB)

56.5/56.5 MB 561.8 MB/s 0:00:00

Collecting nvidia-cusolver-cu12==11.4.5.107 (from torch==2.5.1)

Downloading nvidia_cusolver_cu12-11.4.5.107-py3-none-manylinux1_x86_64.whl (124.2 MB)

124.2/124.2 MB 585.5 MB/s 0:00:00

Collecting nvidia-cuspars-cu12==12.1.0.106 (from torch==2.5.1)

Downloading nvidia_cuspars-cu12-12.1.0.106-py3-none-manylinux1_x86_64.whl (196.0 MB)

196.0/196.0 MB 596.5 MB/s 0:00:00

Collecting nvidia-nccl-cu12==2.21.5 (from torch==2.5.1)

Downloading nvidia_nccl_cu12-2.21.5-py3-none-manylinux2014_x86_64.whl (188.7 MB)

188.7/188.7 MB 600.0 MB/s 0:00:00

Collecting nvidia-nvtx-cu12==12.1.105 (from torch==2.5.1)

Downloading nvidia_nvtx_cu12-12.1.105-py3-none-manylinux1_x86_64.whl (99 kB)

Collecting triton==3.1.0 (from torch==2.5.1)

Downloading triton-3.1.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (209.5 MB)

209.5/209.5 MB 37.4 MB/s 0:00:05

Collecting sympy==1.13.1 (from torch==2.5.1)

Downloading sympy-1.13.1-py3-none-any.whl.metadata (12 kB)

Collecting numpy (from torchvision==0.20.1)

Downloading numpy-2.4.6-cp311-cp311-

manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)

Collecting pillow!=8.3.*,>=5.3.0 (from torchvision==0.20.1)

Downloading pillow-12.3.0-cp311-cp311-

manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (9.1 kB)

Collecting nvidia-nvjitlink-cu12 (from nvidia-cusolver-cu12==11.4.5.107->torch==2.5.1)

Downloading nvidia_nvjitlink_cu12-12.9.86-py3-none-

manylinux2010_x86_64.manylinux_2_12_x86_64.whl (39.7 MB)

39.7/39.7 MB 544.8 MB/s 0:00:00

Collecting mpmath<1.4,>=1.1.0 (from sympy==1.13.1->torch==2.5.1)

Downloading mpmath-1.3.0-py3-none-any.whl.metadata (8.6 kB)

Collecting MarkupSafe>=2.0 (from jinja2->torch==2.5.1)

Downloading markupsafe-3.0.3-cp311-cp311-

manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)

Downloading sympy-1.13.1-py3-none-any.whl (6.2 MB)

6.2/6.2 MB 164.4 MB/s 0:00:00

Downloading mpmath-1.3.0-py3-none-any.whl (536 kB)

536.2/536.2 kB 38.6 MB/s 0:00:00

Downloading pillow-12.3.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (6.9 MB)

6.9/6.9 MB 169.2 MB/s 0:00:00

Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)

Downloading filelock-3.32.3-py3-none-any.whl (98 kB)

Downloading fsspec-2026.7.0-py3-none-any.whl (206 kB)

Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)

Downloading markupsafe-3.0.3-cp311-cp311-

manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)

Downloading networkx-3.6.1-py3-none-any.whl (2.1 MB)

2.1/2.1 MB 143.0 MB/s 0:00:00

Downloading numpy-2.4.6-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (16.9 MB)

16.9/16.9 MB 86.1 MB/s 0:00:00

Installing collected packages: mpmath, typing-extensions, sympy, pillow, nvidia-nvtx-cu12, nvidia-nvjitlink-cu12, nvidia-nccl-cu12, nvidia-curand-cu12, nvidia-cufft-cu12, nvidia-cuda-runtime-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda-cupti-cu12, nvidia-cublas-cu12, numpy, networkx, MarkupSafe, fsspec, filelock, triton, nvidia-cuspars-cu12, nvidia-cudnn-cu12, jinja2, nvidia-cusolver-cu12, torch, torchvision

Successfully installed MarkupSafe-3.0.3 filelock-3.32.3 fsspec-2026.7.0 jinja2-3.1.6 mpmath-1.3.0 networkx-3.6.1 numpy-2.4.6 nvidia-cublas-cu12-12.1.3.1 nvidia-cuda-cupti-cu12-12.1.105 nvidia-cuda-nvrtc-cu12-12.1.105 nvidia-cuda-runtime-cu12-12.1.105 nvidia-cudnn-cu12-9.1.0.70 nvidia-cufft-cu12-11.0.2.54 nvidia-curand-cu12-10.3.2.106 nvidia-cusolver-cu12-11.4.5.107 nvidia-cuspars-cu12-12.1.0.106 nvidia-nccl-cu12-2.21.5 nvidia-nvjitlink-cu12-12.9.86 nvidia-nvtx-cu12-12.1.105 pillow-12.3.0 sympy-1.13.1 torch-2.5.1+cu121 torchvision-0.20.1+cu121 triton-3.1.0 typing-extensions-4.16.0

Collecting transformers

Downloading transformers-5.17.0-py3-none-any.whl.metadata (32 kB)

Collecting accelerate

Downloading accelerate-1.15.0-py3-none-any.whl.metadata (19 kB)

Requirement already satisfied: pillow in ./venv/lib/python3.11/site-packages (12.3.0)

Collecting tqdm

Downloading tqdm-4.70.1-py3-none-any.whl.metadata (57 kB)

Collecting scikit-learn

Downloading scikit_learn-1.9.1-cp311-cp311-

manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (9.3 kB)

Collecting numpy<2

Downloading numpy-1.26.4-cp311-cp311-

manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)

Collecting qwen-vl-utils

Downloading qwen_vl_utils-0.0.14-py3-none-any.whl.metadata (9.0 kB)

Collecting huggingface-hub<2.0,>=1.5.0 (from transformers)

Downloading huggingface_hub-1.32.0-py3-none-any.whl.metadata (16 kB)

Collecting packaging>=20.0 (from transformers)

Downloading packaging-26.3-py3-none-any.whl.metadata (3.5 kB)

Collecting pyyaml>=5.1 (from transformers)

Downloading pyyaml-6.0.3-cp311-cp311-

manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.4 kB)

Collecting regex>=2025.10.22 (from transformers)

Downloading regex-2026.9.10-cp311-cp311-

manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (40 kB)

Collecting tokenizers<0.24.0,>=0.23.1 (from transformers)

Downloading tokenizers-0.23.2-cp310-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (9.8 kB)
Collecting typer (from transformers)
Downloading typer-0.27.2-py3-none-any.whl.metadata (16 kB)
Collecting safetensors>=0.8.0 (from transformers)
Downloading safetensors-0.8.0-cp310-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.2 kB)
Collecting click<9.0.0,>=8.4.2 (from huggingface-hub<2.0,>=1.5.0->transformers)
Downloading click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: filelock>=3.10.0 in ./venv/lib/python3.11/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (3.32.3)
Requirement already satisfied: fsspec>=2023.5.0 in ./venv/lib/python3.11/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (2026.7.0)
Collecting hf-xet<2.0.0,>=1.5.2 (from huggingface-hub<2.0,>=1.5.0->transformers)
Downloading hf_xet-1.6.0-cp38-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (4.9 kB)
Collecting httpx<1,>=0.23.0 (from huggingface-hub<2.0,>=1.5.0->transformers)
Downloading httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: typing-extensions>=4.1.0 in ./venv/lib/python3.11/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (4.16.0)
Collecting anyio (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers)
Downloading anyio-4.15.1-py3-none-any.whl.metadata (4.7 kB)
Collecting certifi (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers)
Downloading certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Collecting httpcore==1.* (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers)
Downloading httpcore-1.0.9-py3-none-any.whl.metadata (21 kB)
Collecting idna (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers)
Downloading idna-3.20-py3-none-any.whl.metadata (7.2 kB)
Collecting h11>=0.16 (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers)
Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting psutil (from accelerate)
Downloading psutil-7.2.2-cp36-abi3-manylinux2010_x86_64.manylinux_2_12_x86_64.manylinux_2_28_x86_64.whl.metadata (22 kB)
Requirement already satisfied: torch>=2.0.0 in ./venv/lib/python3.11/site-packages (from accelerate) (2.5.1+cu121)
Collecting scipy>=1.10.0 (from scikit-learn)
Downloading scipy-1.17.1-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (62 kB)
Collecting joblib>=1.4.0 (from scikit-learn)
Downloading joblib-1.6.0-py3-none-any.whl.metadata (6.5 kB)
Collecting narwhals>=2.0.1 (from scikit-learn)
Downloading narwhals-2.26.0-py3-none-any.whl.metadata (15 kB)
Collecting threadpoolctl>=3.5.0 (from scikit-learn)
Downloading threadpoolctl-3.7.0-py3-none-any.whl.metadata (24 kB)
Collecting av (from qwen-vl-utils)
Downloading av-18.1.0-cp311-abi3-manylinux_2_28_x86_64.whl.metadata (5.0 kB)
Collecting requests (from qwen-vl-utils)
Downloading requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting cloudpickle>=3.0 (from joblib>=1.4.0->scikit-learn)
Downloading cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: networkx in ./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (3.6.1)
Requirement already satisfied: jinja2 in ./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.1.105 in ./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (12.1.105)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.1.105 in ./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (12.1.105)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.1.105 in ./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (12.1.105)
Requirement already satisfied: nvidia-cudnn-cu12==9.1.0.70 in ./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (9.1.0.70)
Requirement already satisfied: nvidia-cublas-cu12==12.1.3.1 in ./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (12.1.3.1)
Requirement already satisfied: nvidia-cufft-cu12==11.0.2.54 in ./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (11.0.2.54)
Requirement already satisfied: nvidia-curand-cu12==10.3.2.106 in

```
./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (10.3.2.106)
Requirement already satisfied: nvidia-cusolver-cu12==11.4.5.107 in
./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (11.4.5.107)
Requirement already satisfied: nvidia-cusparse-cu12==12.1.0.106 in
./venv/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (12.1.0.106)
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in ./venv/lib/python3.11/site-
packages (from torch>=2.0.0->accelerate) (2.21.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.1.105 in ./venv/lib/python3.11/site-
packages (from torch>=2.0.0->accelerate) (12.1.105)
Requirement already satisfied: triton==3.1.0 in ./venv/lib/python3.11/site-packages (from
torch>=2.0.0->accelerate) (3.1.0)
Requirement already satisfied: sympy==1.13.1 in ./venv/lib/python3.11/site-packages (from
torch>=2.0.0->accelerate) (1.13.1)
Requirement already satisfied: nvidia-nvjitlink-cu12 in ./venv/lib/python3.11/site-
packages (from nvidia-cusolver-cu12==11.4.5.107->torch>=2.0.0->accelerate) (12.9.86)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in ./venv/lib/python3.11/site-packages
(from sympy==1.13.1->torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in ./venv/lib/python3.11/site-packages
(from jinja2->torch>=2.0.0->accelerate) (3.0.3)
Collecting charset_normalizer<4,>=2 (from requests->qwen-vl-utils)
  Downloading charset_normalizer-3.5.1-cp311-cp311-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (45 kB)
Collecting urllib3<3,>=1.26 (from requests->qwen-vl-utils)
  Downloading urllib3-2.8.0-py3-none-any.whl.metadata (7.4 kB)
Collecting shellingham>=1.3.0 (from typer->transformers)
  Downloading shellingham-1.5.4-py2.py3-none-any.whl.metadata (3.5 kB)
Collecting rich>=13.8.0 (from typer->transformers)
  Downloading rich-15.0.0-py3-none-any.whl.metadata (18 kB)
Collecting annotated-doc>=0.0.2 (from typer->transformers)
  Downloading annotated_doc-0.0.5-py3-none-any.whl.metadata (6.5 kB)
Collecting markdown-it-py>=2.2.0 (from rich>=13.8.0->typer->transformers)
  Downloading markdown_it_py-4.2.0-py3-none-any.whl.metadata (7.4 kB)
Collecting pygments<3.0.0,>=2.13.0 (from rich>=13.8.0->typer->transformers)
  Downloading pygments-2.21.0-py3-none-any.whl.metadata (2.5 kB)
Collecting mdurl~=0.1 (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers)
  Downloading mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)
Downloading numpy-1.26.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.3
MB)
  _____ 18.3/18.3 MB 58.9 MB/s 0:00:00
Downloading transformers-5.17.0-py3-none-any.whl (12.3 MB)
  _____ 12.3/12.3 MB 185.5 MB/s 0:00:00
Downloading huggingface_hub-1.32.0-py3-none-any.whl (842 kB)
  _____ 842.9/842.9 kB 76.7 MB/s 0:00:00
Downloading click-8.5.0-py3-none-any.whl (125 kB)
Downloading hf_xet-1.6.0-cp38-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (4.5
MB)
  _____ 4.5/4.5 MB 172.3 MB/s 0:00:00
Downloading httpx-0.28.1-py3-none-any.whl (73 kB)
Downloading httpcore-1.0.9-py3-none-any.whl (78 kB)
Downloading tokenizers-0.23.2-cp310-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
(3.4 MB)
  _____ 3.4/3.4 MB 131.2 MB/s 0:00:00
Downloading accelerate-1.15.0-py3-none-any.whl (394 kB)
Downloading tqdm-4.70.1-py3-none-any.whl (80 kB)
Downloading scikit_learn-1.9.1-cp311-cp311-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (9.3 MB)
  _____ 9.3/9.3 MB 208.2 MB/s 0:00:00
Downloading qwen_vl_utils-0.0.14-py3-none-any.whl (8.1 kB)
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Downloading joblib-1.6.0-py3-none-any.whl (306 kB)
Downloading cloudpickle-3.1.2-py3-none-any.whl (22 kB)
Downloading narwhals-2.26.0-py3-none-any.whl (474 kB)
Downloading packaging-26.3-py3-none-any.whl (129 kB)
Downloading pyyaml-6.0.3-cp311-cp311-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (806 kB)
  _____ 806.6/806.6 kB 69.0 MB/s 0:00:00
Downloading regex-2026.9.10-cp311-cp311-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (802 kB)
  _____ 803.0/803.0 kB 75.3 MB/s 0:00:00
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Downloading safetensors-0.8.0-cp310-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (516 kB)

Downloading scipy-1.17.1-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (35.3 MB)

35.3/35.3 MB 50.3 MB/s 0:00:00

Downloading threadpoolctl-3.7.0-py3-none-any.whl (26 kB)

Downloading anyio-4.15.1-py3-none-any.whl (132 kB)

Downloading idna-3.20-py3-none-any.whl (69 kB)

Downloading av-18.1.0-cp311-abi3-manylinux_2_28_x86_64.whl (35.8 MB)

35.8/35.8 MB 121.5 MB/s 0:00:00

Downloading certifi-2026.7.22-py3-none-any.whl (136 kB)

Downloading psutil-7.2.2-cp36-abi3-

manylinux2010_x86_64.manylinux_2_12_x86_64.manylinux_2_28_x86_64.whl (155 kB)

Downloading requests-2.34.2-py3-none-any.whl (73 kB)

Downloading charset_normalizer-3.5.1-cp311-cp311-

manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (262 kB)

Downloading urllib3-2.8.0-py3-none-any.whl (135 kB)

Downloading typer-0.27.2-py3-none-any.whl (123 kB)

Downloading annotated_doc-0.0.5-py3-none-any.whl (5.3 kB)

Downloading rich-15.0.0-py3-none-any.whl (310 kB)

Downloading pygments-2.21.0-py3-none-any.whl (1.3 MB)

1.3/1.3 MB 108.2 MB/s 0:00:00

Downloading markdown_it_py-4.2.0-py3-none-any.whl (91 kB)

Downloading mdurl-0.1.2-py3-none-any.whl (10.0 kB)

Downloading shellingham-1.5.4-py2.py3-none-any.whl (9.8 kB)

Installing collected packages: urllib3, tqdm, threadpoolctl, shellingham, safetensors, regex, pyyaml, pygments, psutil, packaging, numpy, narwhals, mdurl, idna, hf-xet, h11, cloudpickle, click, charset_normalizer, certifi, av, annotated-doc, scipy, requests, markdown-it-py, joblib, httpcore, anyio, scikit-learn, rich, qwen-vl-utils, httpx, typer, huggingface-hub, tokenizers, accelerate, transformers

Attempting uninstall: numpy

Found existing installation: numpy 2.4.6

Uninstalling numpy-2.4.6:

Successfully uninstalled numpy-2.4.6

Successfully installed accelerate-1.15.0 annotated-doc-0.0.5 anyio-4.15.1 av-18.1.0 certifi-2026.7.22 charset_normalizer-3.5.1 click-8.5.0 cloudpickle-3.1.2 h11-0.16.0 hf-xet-1.6.0 httpcore-1.0.9 httpx-0.28.1 huggingface-hub-1.32.0 idna-3.20 joblib-1.6.0 markdown-it-py-4.2.0 mdurl-0.1.2 narwhals-2.26.0 numpy-1.26.4 packaging-26.3 psutil-7.2.2 pygments-2.21.0 pyyaml-6.0.3 qwen-vl-utils-0.0.14 regex-2026.9.10 requests-2.34.2 rich-15.0.0 safetensors-0.8.0 scikit-learn-1.9.1 scipy-1.17.1 shellingham-1.5.4 threadpoolctl-3.7.0 tokenizers-0.23.2 tqdm-4.70.1 transformers-5.17.0 typer-0.27.2 urllib3-2.8.0

host : gcn5.local.snellius.surf.nl

workdir : /gpfs/home6/mdemirev/VisionCCS/vision_ccs

python : /gpfs/home6/mdemirev/VisionCCS/vision_ccs/venv/bin/python

cuda : True NVIDIA A100-SXM4-40GB MIG 3g.20gb

NVIDIA A100-SXM4-40GB, 40960 MiB

NVIDIA A100-SXM4-40GB, 40960 MiB

NVIDIA A100-SXM4-40GB, 40960 MiB

NVIDIA A100-SXM4-40GB, 40960 MiB

=== running linear_ccs.py ===

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

Model: Qwen/Qwen2-VL-7B-Instruct

CATEGORY: OBJECT_DETECTION
#####

=====

EXTRACTING HIDDEN STATES: OBJECT_DETECTION

=====

⚠ No cache found. Starting extraction...

Processing 1306 samples in batches of 8

Searching in 2 image directories

LOADING MODEL: qwen2

Device: cuda

Fetching 5 files:	0%	0/5 [00:00<?, ?it/s]
Fetching 5 files:	20%	1/5 [00:06<00:25, 6.46s/it]
Fetching 5 files:	40%	2/5 [00:16<00:25, 8.36s/it]
Fetching 5 files:	60%	3/5 [00:16<00:09, 4.60s/it]
Fetching 5 files:	80%	4/5 [00:16<00:02, 2.94s/it]
Fetching 5 files:	100%	5/5 [00:16<00:00, 2.00s/it]
Fetching 5 files:	100%	5/5 [00:16<00:00, 3.40s/it]

Loading weights:	0%	0/730 [00:00<?, ?it/s]
Loading weights:	0%	1/730 [00:01<20:09, 1.66s/it]
Loading weights:	0%	2/730 [00:03<20:36, 1.70s/it]
Loading weights:	1%	4/730 [00:03<08:25, 1.43it/s]
Loading weights:	1%	5/730 [00:03<06:34, 1.84it/s]
Loading weights:	1%	6/730 [00:03<05:09, 2.34it/s]
Loading weights:	2%	16/730 [00:04<01:09, 10.20it/s]
Loading weights:	2%	18/730 [00:04<01:17, 9.17it/s]
Loading weights:	4%	28/730 [00:04<00:49, 14.13it/s]
Loading weights:	4%	30/730 [00:05<00:50, 13.93it/s]
Loading weights:	5%	40/730 [00:05<00:32, 21.16it/s]
Loading weights:	6%	43/730 [00:05<00:39, 17.47it/s]
Loading weights:	7%	52/730 [00:05<00:29, 23.04it/s]
Loading weights:	8%	55/730 [00:06<00:36, 18.54it/s]
Loading weights:	9%	64/730 [00:06<00:27, 23.83it/s]
Loading weights:	9%	67/730 [00:06<00:34, 19.07it/s]
Loading weights:	10%	75/730 [00:07<00:33, 19.74it/s]
Loading weights:	11%	78/730 [00:07<00:33, 19.63it/s]
Loading weights:	12%	88/730 [00:07<00:24, 26.10it/s]
Loading weights:	12%	91/730 [00:07<00:31, 20.40it/s]
Loading weights:	14%	100/730 [00:08<00:24, 25.42it/s]
Loading weights:	14%	103/730 [00:08<00:31, 19.93it/s]
Loading weights:	15%	112/730 [00:08<00:24, 24.98it/s]
Loading weights:	16%	115/730 [00:08<00:31, 19.71it/s]
Loading weights:	17%	124/730 [00:09<00:24, 24.84it/s]
Loading weights:	17%	127/730 [00:09<00:30, 19.64it/s]
Loading weights:	19%	136/730 [00:09<00:24, 24.64it/s]
Loading weights:	19%	139/730 [00:09<00:30, 19.44it/s]
Loading weights:	20%	148/730 [00:10<00:23, 24.48it/s]
Loading weights:	21%	151/730 [00:10<00:29, 19.49it/s]
Loading weights:	22%	160/730 [00:10<00:28, 19.90it/s]
Loading weights:	22%	163/730 [00:11<00:28, 19.73it/s]
Loading weights:	24%	172/730 [00:11<00:22, 24.32it/s]
Loading weights:	24%	175/730 [00:11<00:29, 18.88it/s]
Loading weights:	25%	184/730 [00:11<00:22, 24.08it/s]
Loading weights:	26%	187/730 [00:12<00:28, 19.23it/s]
Loading weights:	27%	196/730 [00:12<00:21, 24.44it/s]
Loading weights:	27%	199/730 [00:12<00:27, 19.46it/s]
Loading weights:	28%	208/730 [00:13<00:25, 20.09it/s]
Loading weights:	29%	211/730 [00:13<00:26, 19.93it/s]
Loading weights:	30%	220/730 [00:13<00:25, 19.94it/s]
Loading weights:	31%	223/730 [00:13<00:25, 19.55it/s]
Loading weights:	32%	232/730 [00:14<00:20, 24.60it/s]
Loading weights:	32%	235/730 [00:14<00:25, 19.41it/s]
Loading weights:	33%	244/730 [00:14<00:19, 24.65it/s]
Loading weights:	34%	247/730 [00:15<00:24, 19.58it/s]
Loading weights:	35%	256/730 [00:15<00:19, 24.52it/s]
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Loading weights:	37%	268/730 [00:15<00:19, 24.20it/s]
Loading weights:	37%	271/730 [00:16<00:24, 18.86it/s]
Loading weights:	38%	280/730 [00:16<00:18, 23.93it/s]
Loading weights:	39%	283/730 [00:16<00:23, 19.20it/s]
Loading weights:	40%	292/730 [00:17<00:25, 16.96it/s]
Loading weights:	42%	304/730 [00:17<00:17, 24.18it/s]
Loading weights:	42%	307/730 [00:17<00:21, 20.03it/s]
Loading weights:	43%	316/730 [00:18<00:17, 24.16it/s]
Loading weights:	44%	319/730 [00:18<00:21, 19.21it/s]
Loading weights:	45%	328/730 [00:18<00:16, 23.95it/s]
Loading weights:	45%	331/730 [00:19<00:20, 19.40it/s]
Loading weights:	47%	345/730 [00:19<00:11, 34.53it/s]

Loading weights:	51%		371/730	[00:19<00:05, 67.66it/s]
Loading weights:	54%		395/730	[00:19<00:03, 97.26it/s]
Loading weights:	58%		427/730	[00:19<00:02, 138.77it/s]
Loading weights:	62%		453/730	[00:19<00:01, 160.67it/s]
Loading weights:	66%		479/730	[00:19<00:01, 177.63it/s]
Loading weights:	70%		509/730	[00:19<00:01, 206.87it/s]
Loading weights:	73%		534/730	[00:19<00:00, 216.32it/s]
Loading weights:	77%		560/730	[00:20<00:00, 223.88it/s]
Loading weights:	80%		587/730	[00:20<00:00, 232.05it/s]
Loading weights:	84%		616/730	[00:20<00:00, 239.42it/s]
Loading weights:	88%		645/730	[00:20<00:00, 247.67it/s]
Loading weights:	92%		671/730	[00:20<00:00, 240.97it/s]
Loading weights:	96%		703/730	[00:20<00:00, 256.01it/s]
Loading weights:	100%		729/730	[00:20<00:00, 204.57it/s]
Loading weights:	100%		730/730	[00:20<00:00, 35.18it/s]

✓ Model loaded successfully

Batches:	0%		0/164	[00:00<?, ?it/s]
Batches:	1%		1/164	[00:03<09:52, 3.64s/it]
Batches:	1%		2/164	[00:06<07:53, 2.92s/it]
Batches:	2%		3/164	[00:08<07:00, 2.61s/it]
Batches:	2%		4/164	[00:10<06:36, 2.48s/it]
Batches:	3%		5/164	[00:13<06:32, 2.47s/it]
Batches:	4%		6/164	[00:15<06:23, 2.43s/it]
Batches:	4%		7/164	[00:17<06:16, 2.40s/it]
Batches:	5%		8/164	[00:20<06:12, 2.39s/it]
Batches:	5%		9/164	[00:22<06:18, 2.44s/it]
Batches:	6%		10/164	[00:24<06:01, 2.35s/it]
Batches:	7%		11/164	[00:27<05:53, 2.31s/it]
Batches:	7%		12/164	[00:29<05:53, 2.33s/it]
Batches:	8%		13/164	[00:31<05:54, 2.34s/it]
Batches:	9%		14/164	[00:34<05:58, 2.39s/it]
Batches:	9%		15/164	[00:36<05:57, 2.40s/it]
Batches:	10%		16/164	[00:39<06:10, 2.50s/it]
Batches:	10%		17/164	[00:41<06:07, 2.50s/it]
Batches:	11%		18/164	[00:44<06:03, 2.49s/it]
Batches:	12%		19/164	[00:46<05:58, 2.47s/it]
Batches:	12%		20/164	[00:49<05:54, 2.46s/it]
Batches:	13%		21/164	[00:51<05:45, 2.42s/it]
Batches:	13%		22/164	[00:54<05:45, 2.44s/it]
Batches:	14%		23/164	[00:56<05:45, 2.45s/it]
Batches:	15%		24/164	[00:58<05:37, 2.41s/it]
Batches:	15%		25/164	[01:01<05:32, 2.39s/it]
Batches:	16%		26/164	[01:03<05:25, 2.36s/it]
Batches:	16%		27/164	[01:06<05:31, 2.42s/it]
Batches:	17%		28/164	[01:08<05:29, 2.43s/it]
Batches:	18%		29/164	[01:10<05:19, 2.36s/it]
Batches:	18%		30/164	[01:13<05:18, 2.38s/it]
Batches:	19%		31/164	[01:15<05:19, 2.40s/it]
Batches:	20%		32/164	[01:17<05:13, 2.38s/it]
Batches:	20%		33/164	[01:20<05:18, 2.43s/it]
Batches:	21%		34/164	[01:22<05:14, 2.42s/it]
Batches:	21%		35/164	[01:25<05:10, 2.41s/it]
Batches:	22%		36/164	[01:27<05:05, 2.39s/it]
Batches:	23%		37/164	[01:30<05:14, 2.48s/it]
Batches:	23%		38/164	[01:32<04:59, 2.37s/it]
Batches:	24%		39/164	[01:34<04:55, 2.36s/it]
Batches:	24%		40/164	[01:36<04:42, 2.28s/it]
Batches:	25%		41/164	[01:39<04:50, 2.36s/it]
Batches:	26%		42/164	[01:41<04:43, 2.32s/it]
Batches:	26%		43/164	[01:44<04:45, 2.36s/it]
Batches:	27%		44/164	[01:46<04:44, 2.37s/it]
Batches:	27%		45/164	[01:48<04:42, 2.38s/it]
Batches:	28%		46/164	[01:51<04:41, 2.39s/it]
Batches:	29%		47/164	[01:53<04:32, 2.33s/it]
Batches:	29%		48/164	[01:55<04:35, 2.37s/it]
Batches:	30%		49/164	[01:58<04:29, 2.34s/it]
Batches:	30%		50/164	[02:00<04:22, 2.31s/it]

Batches: 31%	51/164	[02:02<04:24, 2.34s/it]
Batches: 32%	52/164	[02:05<04:23, 2.35s/it]
Batches: 32%	53/164	[02:07<04:17, 2.32s/it]
Batches: 33%	54/164	[02:10<04:25, 2.41s/it]
Batches: 34%	55/164	[02:12<04:17, 2.36s/it]
Batches: 34%	56/164	[02:14<04:08, 2.30s/it]
Batches: 35%	57/164	[02:16<04:08, 2.32s/it]
Batches: 35%	58/164	[02:19<04:06, 2.32s/it]
Batches: 36%	59/164	[02:21<04:02, 2.31s/it]
Batches: 37%	60/164	[02:23<04:07, 2.38s/it]
Batches: 37%	61/164	[02:26<03:59, 2.32s/it]
Batches: 38%	62/164	[02:28<03:56, 2.31s/it]
Batches: 38%	63/164	[02:30<03:49, 2.27s/it]
Batches: 39%	64/164	[02:32<03:48, 2.28s/it]
Batches: 40%	65/164	[02:35<03:53, 2.36s/it]
Batches: 40%	66/164	[02:37<03:47, 2.32s/it]
Batches: 41%	67/164	[02:39<03:43, 2.31s/it]
Batches: 41%	68/164	[02:42<03:44, 2.34s/it]
Batches: 42%	69/164	[02:44<03:44, 2.36s/it]
Batches: 43%	70/164	[02:47<03:43, 2.38s/it]
Batches: 43%	71/164	[02:49<03:42, 2.40s/it]
Batches: 44%	72/164	[02:51<03:38, 2.37s/it]
Batches: 45%	73/164	[02:54<03:31, 2.32s/it]
Batches: 45%	74/164	[02:56<03:24, 2.27s/it]
Batches: 46%	75/164	[02:58<03:23, 2.29s/it]
Batches: 46%	76/164	[03:00<03:17, 2.25s/it]
Batches: 47%	77/164	[03:03<03:18, 2.28s/it]
Batches: 48%	78/164	[03:05<03:16, 2.29s/it]
Batches: 48%	79/164	[03:07<03:11, 2.25s/it]
Batches: 49%	80/164	[03:09<03:11, 2.27s/it]
Batches: 49%	81/164	[03:12<03:06, 2.25s/it]
Batches: 50%	82/164	[03:14<03:06, 2.28s/it]
Batches: 51%	83/164	[03:16<03:06, 2.31s/it]
Batches: 51%	84/164	[03:19<03:01, 2.27s/it]
Batches: 52%	85/164	[03:21<02:56, 2.23s/it]
Batches: 52%	86/164	[03:23<02:52, 2.21s/it]
Batches: 53%	87/164	[03:25<02:59, 2.33s/it]
Batches: 54%	88/164	[03:28<02:57, 2.34s/it]
Batches: 54%	89/164	[03:30<02:52, 2.30s/it]
Batches: 55%	90/164	[03:32<02:46, 2.25s/it]
Batches: 55%	91/164	[03:35<02:46, 2.28s/it]
Batches: 56%	92/164	[03:37<02:45, 2.31s/it]
Batches: 57%	93/164	[03:39<02:44, 2.31s/it]
Batches: 57%	94/164	[03:42<02:44, 2.34s/it]
Batches: 58%	95/164	[03:44<02:51, 2.49s/it]
Batches: 59%	96/164	[03:47<02:46, 2.46s/it]
Batches: 59%	97/164	[03:49<02:45, 2.47s/it]
Batches: 60%	98/164	[03:52<02:41, 2.45s/it]
Batches: 60%	99/164	[03:54<02:33, 2.37s/it]
Batches: 61%	100/164	[03:56<02:31, 2.36s/it]
Batches: 62%	101/164	[03:59<02:27, 2.34s/it]
Batches: 62%	102/164	[04:01<02:23, 2.31s/it]
Batches: 63%	103/164	[04:03<02:23, 2.35s/it]
Batches: 63%	104/164	[04:06<02:19, 2.33s/it]
Batches: 64%	105/164	[04:08<02:18, 2.34s/it]
Batches: 65%	106/164	[04:10<02:15, 2.33s/it]
Batches: 65%	107/164	[04:13<02:14, 2.37s/it]
Batches: 66%	108/164	[04:15<02:07, 2.28s/it]
Batches: 66%	109/164	[04:17<02:05, 2.29s/it]
Batches: 67%	110/164	[04:20<02:10, 2.42s/it]
Batches: 68%	111/164	[04:22<02:07, 2.41s/it]
Batches: 68%	112/164	[04:24<02:02, 2.36s/it]
Batches: 69%	113/164	[04:27<02:00, 2.37s/it]
Batches: 70%	114/164	[04:29<01:58, 2.37s/it]
Batches: 70%	115/164	[04:32<01:56, 2.37s/it]
Batches: 71%	116/164	[04:34<01:51, 2.33s/it]
Batches: 71%	117/164	[04:36<01:47, 2.29s/it]
Batches: 72%	118/164	[04:38<01:45, 2.28s/it]
Batches: 73%	119/164	[04:40<01:41, 2.26s/it]

Batches: 73%		120/164	[04:43<01:41, 2.31s/it]
Batches: 74%		121/164	[04:45<01:40, 2.34s/it]
Batches: 74%		122/164	[04:47<01:35, 2.27s/it]
Batches: 75%		123/164	[04:50<01:34, 2.30s/it]
Batches: 76%		124/164	[04:52<01:32, 2.32s/it]
Batches: 76%		125/164	[04:55<01:31, 2.34s/it]
Batches: 77%		126/164	[04:57<01:28, 2.34s/it]
Batches: 77%		127/164	[04:59<01:27, 2.36s/it]
Batches: 78%		128/164	[05:02<01:24, 2.34s/it]
Batches: 79%		129/164	[05:04<01:18, 2.25s/it]
Batches: 79%		130/164	[05:06<01:17, 2.28s/it]
Batches: 80%		131/164	[05:08<01:16, 2.32s/it]
Batches: 80%		132/164	[05:11<01:12, 2.27s/it]
Batches: 81%		133/164	[05:13<01:11, 2.32s/it]
Batches: 82%		134/164	[05:15<01:08, 2.29s/it]
Batches: 82%		135/164	[05:18<01:07, 2.34s/it]
Batches: 83%		136/164	[05:20<01:06, 2.37s/it]
Batches: 84%		137/164	[05:23<01:04, 2.39s/it]
Batches: 84%		138/164	[05:25<00:59, 2.30s/it]
Batches: 85%		139/164	[05:27<00:58, 2.34s/it]
Batches: 85%		140/164	[05:29<00:55, 2.32s/it]
Batches: 86%		141/164	[05:32<00:53, 2.33s/it]
Batches: 87%		142/164	[05:34<00:50, 2.31s/it]
Batches: 87%		143/164	[05:36<00:49, 2.35s/it]
Batches: 88%		144/164	[05:39<00:46, 2.34s/it]
Batches: 88%		145/164	[05:41<00:46, 2.45s/it]
Batches: 89%		146/164	[05:44<00:43, 2.40s/it]
Batches: 90%		147/164	[05:46<00:40, 2.39s/it]
Batches: 90%		148/164	[05:48<00:37, 2.34s/it]
Batches: 91%		149/164	[05:51<00:35, 2.39s/it]
Batches: 91%		150/164	[05:53<00:32, 2.34s/it]
Batches: 92%		151/164	[05:56<00:31, 2.41s/it]
Batches: 93%		152/164	[05:58<00:28, 2.41s/it]
Batches: 93%		153/164	[06:01<00:27, 2.47s/it]
Batches: 94%		154/164	[06:03<00:24, 2.44s/it]
Batches: 95%		155/164	[06:05<00:22, 2.45s/it]
Batches: 95%		156/164	[06:08<00:19, 2.39s/it]
Batches: 96%		157/164	[06:10<00:16, 2.37s/it]
Batches: 96%		158/164	[06:12<00:14, 2.35s/it]
Batches: 97%		159/164	[06:15<00:11, 2.37s/it]
Batches: 98%		160/164	[06:17<00:09, 2.33s/it]
Batches: 98%		161/164	[06:19<00:07, 2.34s/it]
Batches: 99%		162/164	[06:22<00:04, 2.39s/it]
Batches: 99%		163/164	[06:24<00:02, 2.39s/it]
Batches: 100%		164/164	[06:25<00:00, 1.89s/it]
Batches: 100%		164/164	[06:25<00:00, 2.35s/it]

=====

✓ Successfully processed: 1306/1306

x Skipped (missing/error): 0/1306

Extracted shapes:

Positive: (1306, 3584)

Negative: (1306, 3584)

Labels: (1306,)

Cached to: hidden_states_cache_final/cache_object_detection_1306_qwen2_ccs_aligned.npz

Logistic regression sanity-check accuracy: 82.4%

=====

TRAINING CCS PROBE

=====

Dataset split:

Train: 783 samples (402 pos, 381 neg)

Test: 523 samples (251 pos, 272 neg)

Hidden dim: 3584

=====

TRAINING WITH MULTIPLE RANDOM RESTARTS

=====

Trial 1/10: Loss = 0.010284
Trial 2/10: Loss = 0.003833
Trial 3/10: Loss = 0.001281
Trial 4/10: Loss = 0.006388
Trial 5/10: Loss = 0.001281
Trial 6/10: Loss = 0.003834
Trial 7/10: Loss = 0.005112
Trial 8/10: Loss = 0.011503
Trial 9/10: Loss = 0.006390
Trial 10/10: Loss = 0.011496

=====

EVALUATION WITH BEST PROBE

=====

Best loss: 0.001281

Test Results:

Overall Accuracy: 89.9% (470/523)
Positive samples: 94.0% (251 samples)
Negative samples: 86.0% (272 samples)

✓ COMPLETE: object_detection → 89.9%

CATEGORY: ATTRIBUTE_RECOGNITION
#####

=====

EXTRACTING HIDDEN STATES: ATTRIBUTE_RECOGNITION

=====

⚠ No cache found. Starting extraction...

Processing 3366 samples in batches of 8
Searching in 2 image directories
LOADING MODEL: qwen2
Device: cuda

Fetching 5 files: 0% | 0/5 [00:00<?, ?it/s]
Fetching 5 files: 100% | 5/5 [00:00<00:00, 1401.65it/s]

Loading weights: 0% | 0/730 [00:00<?, ?it/s]
Loading weights: 0% | 1/730 [00:00<05:30, 2.21it/s]
Loading weights: 0% | 2/730 [00:01<06:10, 1.97it/s]
Loading weights: 1% | 5/730 [00:01<02:07, 5.67it/s]
Loading weights: 2% | 16/730 [00:01<00:34, 20.47it/s]
Loading weights: 3% | 20/730 [00:01<00:31, 22.55it/s]
Loading weights: 4% | 28/730 [00:01<00:23, 30.05it/s]
Loading weights: 5% | 40/730 [00:01<00:16, 42.74it/s]
Loading weights: 6% | 46/730 [00:01<00:16, 42.31it/s]
Loading weights: 7% | 51/730 [00:02<00:16, 40.08it/s]
Loading weights: 9% | 64/730 [00:02<00:12, 52.43it/s]
Loading weights: 10% | 70/730 [00:02<00:13, 49.13it/s]
Loading weights: 10% | 76/730 [00:02<00:13, 46.72it/s]
Loading weights: 12% | 88/730 [00:02<00:11, 55.34it/s]
Loading weights: 13% | 94/730 [00:02<00:12, 51.18it/s]
Loading weights: 14% | 100/730 [00:02<00:13, 48.04it/s]
Loading weights: 15% | 112/730 [00:03<00:10, 56.25it/s]
Loading weights: 16% | 118/730 [00:03<00:11, 51.81it/s]
Loading weights: 17% | 125/730 [00:03<00:13, 44.72it/s]
Loading weights: 19% | 136/730 [00:03<00:11, 51.53it/s]
Loading weights: 20% | 148/730 [00:03<00:09, 58.21it/s]
Loading weights: 21% | 154/730 [00:03<00:10, 53.43it/s]
Loading weights: 22% | 161/730 [00:04<00:10, 51.79it/s]
Loading weights: 24% | 172/730 [00:04<00:11, 49.17it/s]
Loading weights: 25% | 180/730 [00:04<00:10, 54.99it/s]
Loading weights: 25% | 186/730 [00:04<00:11, 46.47it/s]

Loading weights:	27%	197/730	[00:04<00:10, 53.14it/s]
Loading weights:	28%	208/730	[00:05<00:10, 51.87it/s]
Loading weights:	30%	220/730	[00:05<00:08, 57.59it/s]
Loading weights:	31%	226/730	[00:05<00:09, 51.99it/s]
Loading weights:	32%	232/730	[00:05<00:10, 49.35it/s]
Loading weights:	33%	244/730	[00:05<00:09, 51.09it/s]
Loading weights:	35%	256/730	[00:05<00:08, 58.01it/s]
Loading weights:	36%	262/730	[00:05<00:08, 53.87it/s]
Loading weights:	37%	269/730	[00:06<00:09, 50.84it/s]
Loading weights:	38%	276/730	[00:06<00:08, 54.80it/s]
Loading weights:	39%	282/730	[00:06<00:09, 46.02it/s]
Loading weights:	40%	292/730	[00:06<00:08, 51.40it/s]
Loading weights:	42%	304/730	[00:06<00:08, 52.58it/s]
Loading weights:	43%	316/730	[00:06<00:07, 58.20it/s]
Loading weights:	44%	322/730	[00:07<00:07, 52.29it/s]
Loading weights:	45%	328/730	[00:07<00:08, 49.56it/s]
Loading weights:	47%	344/730	[00:07<00:05, 72.09it/s]
Loading weights:	56%	407/730	[00:07<00:01, 197.92it/s]
Loading weights:	65%	475/730	[00:07<00:00, 312.98it/s]
Loading weights:	74%	537/730	[00:07<00:00, 388.55it/s]
Loading weights:	82%	599/730	[00:07<00:00, 446.00it/s]
Loading weights:	91%	667/730	[00:07<00:00, 505.46it/s]
Loading weights:	99%	725/730	[00:08<00:00, 503.93it/s]
Loading weights:	100%	730/730	[00:08<00:00, 90.60it/s]

✓ Model loaded successfully

Batches:	0%	0/421	[00:00<?, ?it/s]
Batches:	0%	1/421	[00:02<17:04, 2.44s/it]
Batches:	0%	2/421	[00:04<15:27, 2.21s/it]
Batches:	1%	3/421	[00:06<16:09, 2.32s/it]
Batches:	1%	4/421	[00:09<16:32, 2.38s/it]
Batches:	1%	5/421	[00:11<16:12, 2.34s/it]
Batches:	1%	6/421	[00:14<16:30, 2.39s/it]
Batches:	2%	7/421	[00:16<16:03, 2.33s/it]
Batches:	2%	8/421	[00:18<15:34, 2.26s/it]
Batches:	2%	9/421	[00:20<15:44, 2.29s/it]
Batches:	2%	10/421	[00:23<15:58, 2.33s/it]
Batches:	3%	11/421	[00:25<15:43, 2.30s/it]
Batches:	3%	12/421	[00:27<15:32, 2.28s/it]
Batches:	3%	13/421	[00:30<15:55, 2.34s/it]
Batches:	3%	14/421	[00:32<16:21, 2.41s/it]
Batches:	4%	15/421	[00:35<16:20, 2.41s/it]
Batches:	4%	16/421	[00:37<16:41, 2.47s/it]
Batches:	4%	17/421	[00:40<16:49, 2.50s/it]
Batches:	4%	18/421	[00:42<16:27, 2.45s/it]
Batches:	5%	19/421	[00:45<16:19, 2.44s/it]
Batches:	5%	20/421	[00:47<16:15, 2.43s/it]
Batches:	5%	21/421	[00:49<15:29, 2.32s/it]
Batches:	5%	22/421	[00:51<15:30, 2.33s/it]
Batches:	5%	23/421	[00:54<15:49, 2.38s/it]
Batches:	6%	24/421	[00:56<15:51, 2.40s/it]
Batches:	6%	25/421	[00:59<15:39, 2.37s/it]
Batches:	6%	26/421	[01:01<15:36, 2.37s/it]
Batches:	6%	27/421	[01:03<15:37, 2.38s/it]
Batches:	7%	28/421	[01:06<16:03, 2.45s/it]
Batches:	7%	29/421	[01:09<16:21, 2.50s/it]
Batches:	7%	30/421	[01:11<16:28, 2.53s/it]
Batches:	7%	31/421	[01:14<16:27, 2.53s/it]
Batches:	8%	32/421	[01:16<16:17, 2.51s/it]
Batches:	8%	33/421	[01:19<15:59, 2.47s/it]
Batches:	8%	34/421	[01:21<15:52, 2.46s/it]
Batches:	8%	35/421	[01:23<15:13, 2.37s/it]
Batches:	9%	36/421	[01:26<15:01, 2.34s/it]
Batches:	9%	37/421	[01:28<15:02, 2.35s/it]
Batches:	9%	38/421	[01:30<14:52, 2.33s/it]
Batches:	9%	39/421	[01:33<14:59, 2.35s/it]
Batches:	10%	40/421	[01:35<14:56, 2.35s/it]
Batches:	10%	41/421	[01:38<15:20, 2.42s/it]

Batches: 10%	42/421	[01:40<14:50, 2.35s/it]
Batches: 10%	43/421	[01:42<14:33, 2.31s/it]
Batches: 10%	44/421	[01:44<14:40, 2.34s/it]
Batches: 11%	45/421	[01:47<14:26, 2.30s/it]
Batches: 11%	46/421	[01:49<14:15, 2.28s/it]
Batches: 11%	47/421	[01:51<14:38, 2.35s/it]
Batches: 11%	48/421	[01:54<14:29, 2.33s/it]
Batches: 12%	49/421	[01:56<14:31, 2.34s/it]
Batches: 12%	50/421	[01:58<14:48, 2.40s/it]
Batches: 12%	51/421	[02:01<14:57, 2.43s/it]
Batches: 12%	52/421	[02:03<14:56, 2.43s/it]
Batches: 13%	53/421	[02:06<14:53, 2.43s/it]
Batches: 13%	54/421	[02:08<14:20, 2.35s/it]
Batches: 13%	55/421	[02:10<14:30, 2.38s/it]
Batches: 13%	56/421	[02:13<14:17, 2.35s/it]
Batches: 14%	57/421	[02:15<14:33, 2.40s/it]
Batches: 14%	58/421	[02:17<13:53, 2.30s/it]
Batches: 14%	59/421	[02:20<13:54, 2.30s/it]
Batches: 14%	60/421	[02:22<14:03, 2.34s/it]
Batches: 14%	61/421	[02:24<14:11, 2.37s/it]
Batches: 15%	62/421	[02:27<13:53, 2.32s/it]
Batches: 15%	63/421	[02:29<13:48, 2.31s/it]
Batches: 15%	64/421	[02:31<13:35, 2.28s/it]
Batches: 15%	65/421	[02:34<13:46, 2.32s/it]
Batches: 16%	66/421	[02:36<13:48, 2.33s/it]
Batches: 16%	67/421	[02:39<14:14, 2.41s/it]
Batches: 16%	68/421	[02:41<14:18, 2.43s/it]
Batches: 16%	69/421	[02:43<13:59, 2.39s/it]
Batches: 17%	70/421	[02:46<14:03, 2.40s/it]
Batches: 17%	71/421	[02:48<14:03, 2.41s/it]
Batches: 17%	72/421	[02:51<14:18, 2.46s/it]
Batches: 17%	73/421	[02:53<14:09, 2.44s/it]
Batches: 18%	74/421	[02:55<13:21, 2.31s/it]
Batches: 18%	75/421	[02:58<13:23, 2.32s/it]
Batches: 18%	76/421	[03:00<13:22, 2.32s/it]
Batches: 18%	77/421	[03:03<13:54, 2.43s/it]
Batches: 19%	78/421	[03:05<13:43, 2.40s/it]
Batches: 19%	79/421	[03:07<13:21, 2.34s/it]
Batches: 19%	80/421	[03:09<12:56, 2.28s/it]
Batches: 19%	81/421	[03:11<12:45, 2.25s/it]
Batches: 19%	82/421	[03:14<13:01, 2.31s/it]
Batches: 20%	83/421	[03:16<13:08, 2.33s/it]
Batches: 20%	84/421	[03:18<12:34, 2.24s/it]
Batches: 20%	85/421	[03:21<12:55, 2.31s/it]
Batches: 20%	86/421	[03:23<12:55, 2.31s/it]
Batches: 21%	87/421	[03:26<13:11, 2.37s/it]
Batches: 21%	88/421	[03:28<13:05, 2.36s/it]
Batches: 21%	89/421	[03:30<12:55, 2.34s/it]
Batches: 21%	90/421	[03:32<12:43, 2.31s/it]
Batches: 22%	91/421	[03:35<12:44, 2.32s/it]
Batches: 22%	92/421	[03:37<12:29, 2.28s/it]
Batches: 22%	93/421	[03:39<12:30, 2.29s/it]
Batches: 22%	94/421	[03:42<12:41, 2.33s/it]
Batches: 23%	95/421	[03:44<12:38, 2.33s/it]
Batches: 23%	96/421	[03:46<12:32, 2.31s/it]
Batches: 23%	97/421	[03:49<12:32, 2.32s/it]
Batches: 23%	98/421	[03:51<12:21, 2.30s/it]
Batches: 24%	99/421	[03:53<12:12, 2.28s/it]
Batches: 24%	100/421	[03:55<12:14, 2.29s/it]
Batches: 24%	101/421	[03:58<12:38, 2.37s/it]
Batches: 24%	102/421	[04:00<12:25, 2.34s/it]
Batches: 24%	103/421	[04:02<12:15, 2.31s/it]
Batches: 25%	104/421	[04:05<12:16, 2.32s/it]
Batches: 25%	105/421	[04:07<11:52, 2.25s/it]
Batches: 25%	106/421	[04:09<11:49, 2.25s/it]
Batches: 25%	107/421	[04:11<11:55, 2.28s/it]
Batches: 26%	108/421	[04:14<11:40, 2.24s/it]
Batches: 26%	109/421	[04:16<11:56, 2.30s/it]
Batches: 26%	110/421	[04:18<11:58, 2.31s/it]

Batches: 26%	111/421	[04:21<12:08, 2.35s/it]
Batches: 27%	112/421	[04:23<12:10, 2.36s/it]
Batches: 27%	113/421	[04:26<12:09, 2.37s/it]
Batches: 27%	114/421	[04:28<11:53, 2.32s/it]
Batches: 27%	115/421	[04:30<11:42, 2.30s/it]
Batches: 28%	116/421	[04:33<11:55, 2.35s/it]
Batches: 28%	117/421	[04:35<11:54, 2.35s/it]
Batches: 28%	118/421	[04:37<12:03, 2.39s/it]
Batches: 28%	119/421	[04:40<11:59, 2.38s/it]
Batches: 29%	120/421	[04:42<11:41, 2.33s/it]
Batches: 29%	121/421	[04:44<11:47, 2.36s/it]
Batches: 29%	122/421	[04:47<11:58, 2.40s/it]
Batches: 29%	123/421	[04:49<12:02, 2.43s/it]
Batches: 29%	124/421	[04:52<11:43, 2.37s/it]
Batches: 30%	125/421	[04:54<11:34, 2.35s/it]
Batches: 30%	126/421	[04:56<11:33, 2.35s/it]
Batches: 30%	127/421	[04:59<11:45, 2.40s/it]
Batches: 30%	128/421	[05:01<11:39, 2.39s/it]
Batches: 31%	129/421	[05:04<12:04, 2.48s/it]
Batches: 31%	130/421	[05:06<12:00, 2.48s/it]
Batches: 31%	131/421	[05:09<12:03, 2.50s/it]
Batches: 31%	132/421	[05:11<11:41, 2.43s/it]
Batches: 32%	133/421	[05:13<11:30, 2.40s/it]
Batches: 32%	134/421	[05:16<11:00, 2.30s/it]
Batches: 32%	135/421	[05:18<11:07, 2.33s/it]
Batches: 32%	136/421	[05:20<11:11, 2.35s/it]
Batches: 33%	137/421	[05:23<11:18, 2.39s/it]
Batches: 33%	138/421	[05:25<11:24, 2.42s/it]
Batches: 33%	139/421	[05:28<11:25, 2.43s/it]
Batches: 33%	140/421	[05:30<11:34, 2.47s/it]
Batches: 33%	141/421	[05:33<11:32, 2.47s/it]
Batches: 34%	142/421	[05:35<11:22, 2.45s/it]
Batches: 34%	143/421	[05:37<11:06, 2.40s/it]
Batches: 34%	144/421	[05:40<10:52, 2.36s/it]
Batches: 34%	145/421	[05:42<11:06, 2.42s/it]
Batches: 35%	146/421	[05:45<11:07, 2.43s/it]
Batches: 35%	147/421	[05:47<10:56, 2.40s/it]
Batches: 35%	148/421	[05:50<11:00, 2.42s/it]
Batches: 35%	149/421	[05:52<10:50, 2.39s/it]
Batches: 36%	150/421	[05:54<11:05, 2.46s/it]
Batches: 36%	151/421	[05:57<10:57, 2.44s/it]
Batches: 36%	152/421	[05:59<10:51, 2.42s/it]
Batches: 36%	153/421	[06:02<10:46, 2.41s/it]
Batches: 37%	154/421	[06:04<10:38, 2.39s/it]
Batches: 37%	155/421	[06:06<10:28, 2.36s/it]
Batches: 37%	156/421	[06:08<10:11, 2.31s/it]
Batches: 37%	157/421	[06:11<10:11, 2.31s/it]
Batches: 38%	158/421	[06:13<10:07, 2.31s/it]
Batches: 38%	159/421	[06:15<09:57, 2.28s/it]
Batches: 38%	160/421	[06:18<10:00, 2.30s/it]
Batches: 38%	161/421	[06:20<10:05, 2.33s/it]
Batches: 38%	162/421	[06:22<10:12, 2.36s/it]
Batches: 39%	163/421	[06:25<09:51, 2.29s/it]
Batches: 39%	164/421	[06:27<09:56, 2.32s/it]
Batches: 39%	165/421	[06:29<10:02, 2.35s/it]
Batches: 39%	166/421	[06:32<09:53, 2.33s/it]
Batches: 40%	167/421	[06:34<10:17, 2.43s/it]
Batches: 40%	168/421	[06:37<10:14, 2.43s/it]
Batches: 40%	169/421	[06:39<09:55, 2.36s/it]
Batches: 40%	170/421	[06:41<09:42, 2.32s/it]
Batches: 41%	171/421	[06:44<09:43, 2.33s/it]
Batches: 41%	172/421	[06:46<09:45, 2.35s/it]
Batches: 41%	173/421	[06:48<09:47, 2.37s/it]
Batches: 41%	174/421	[06:51<09:36, 2.33s/it]
Batches: 42%	175/421	[06:53<09:40, 2.36s/it]
Batches: 42%	176/421	[06:55<09:39, 2.36s/it]
Batches: 42%	177/421	[06:58<09:25, 2.32s/it]
Batches: 42%	178/421	[07:00<09:34, 2.37s/it]
Batches: 43%	179/421	[07:02<09:26, 2.34s/it]

Batches: 43%	180/421	[07:05<09:29, 2.36s/it]
Batches: 43%	181/421	[07:07<09:32, 2.39s/it]
Batches: 43%	182/421	[07:09<09:14, 2.32s/it]
Batches: 43%	183/421	[07:12<09:12, 2.32s/it]
Batches: 44%	184/421	[07:14<09:12, 2.33s/it]
Batches: 44%	185/421	[07:16<08:58, 2.28s/it]
Batches: 44%	186/421	[07:18<08:42, 2.22s/it]
Batches: 44%	187/421	[07:21<08:49, 2.26s/it]
Batches: 45%	188/421	[07:23<08:54, 2.29s/it]
Batches: 45%	189/421	[07:25<08:32, 2.21s/it]
Batches: 45%	190/421	[07:28<08:49, 2.29s/it]
Batches: 45%	191/421	[07:30<08:51, 2.31s/it]
Batches: 46%	192/421	[07:32<08:58, 2.35s/it]
Batches: 46%	193/421	[07:35<08:54, 2.34s/it]
Batches: 46%	194/421	[07:37<08:45, 2.31s/it]
Batches: 46%	195/421	[07:39<08:50, 2.35s/it]
Batches: 47%	196/421	[07:42<08:38, 2.31s/it]
Batches: 47%	197/421	[07:44<08:38, 2.31s/it]
Batches: 47%	198/421	[07:46<08:31, 2.30s/it]
Batches: 47%	199/421	[07:48<08:22, 2.26s/it]
Batches: 48%	200/421	[07:51<08:18, 2.26s/it]
Batches: 48%	201/421	[07:53<08:12, 2.24s/it]
Batches: 48%	202/421	[07:55<08:28, 2.32s/it]
Batches: 48%	203/421	[07:58<08:31, 2.35s/it]
Batches: 48%	204/421	[08:00<08:25, 2.33s/it]
Batches: 49%	205/421	[08:02<08:13, 2.29s/it]
Batches: 49%	206/421	[08:04<08:14, 2.30s/it]
Batches: 49%	207/421	[08:07<08:20, 2.34s/it]
Batches: 49%	208/421	[08:09<08:09, 2.30s/it]
Batches: 50%	209/421	[08:11<08:09, 2.31s/it]
Batches: 50%	210/421	[08:14<08:01, 2.28s/it]
Batches: 50%	211/421	[08:16<08:06, 2.32s/it]
Batches: 50%	212/421	[08:18<08:07, 2.33s/it]
Batches: 51%	213/421	[08:21<08:22, 2.42s/it]
Batches: 51%	214/421	[08:23<07:48, 2.26s/it]
Batches: 51%	215/421	[08:25<07:57, 2.32s/it]
Batches: 51%	216/421	[08:28<08:09, 2.39s/it]
Batches: 52%	217/421	[08:30<07:53, 2.32s/it]
Batches: 52%	218/421	[08:33<07:54, 2.34s/it]
Batches: 52%	219/421	[08:35<07:45, 2.30s/it]
Batches: 52%	220/421	[08:37<07:45, 2.32s/it]
Batches: 52%	221/421	[08:40<07:53, 2.37s/it]
Batches: 53%	222/421	[08:42<07:51, 2.37s/it]
Batches: 53%	223/421	[08:44<07:58, 2.42s/it]
Batches: 53%	224/421	[08:47<08:01, 2.44s/it]
Batches: 53%	225/421	[08:50<08:06, 2.48s/it]
Batches: 54%	226/421	[08:52<08:03, 2.48s/it]
Batches: 54%	227/421	[08:54<07:57, 2.46s/it]
Batches: 54%	228/421	[08:57<07:45, 2.41s/it]
Batches: 54%	229/421	[08:59<07:26, 2.32s/it]
Batches: 55%	230/421	[09:01<07:22, 2.32s/it]
Batches: 55%	231/421	[09:03<07:08, 2.26s/it]
Batches: 55%	232/421	[09:06<07:21, 2.33s/it]
Batches: 55%	233/421	[09:08<07:18, 2.33s/it]
Batches: 56%	234/421	[09:11<07:20, 2.36s/it]
Batches: 56%	235/421	[09:13<07:24, 2.39s/it]
Batches: 56%	236/421	[09:15<07:24, 2.40s/it]
Batches: 56%	237/421	[09:18<07:08, 2.33s/it]
Batches: 57%	238/421	[09:20<07:05, 2.33s/it]
Batches: 57%	239/421	[09:22<07:10, 2.36s/it]
Batches: 57%	240/421	[09:25<07:05, 2.35s/it]
Batches: 57%	241/421	[09:27<07:09, 2.38s/it]
Batches: 57%	242/421	[09:30<07:06, 2.38s/it]
Batches: 58%	243/421	[09:32<07:10, 2.42s/it]
Batches: 58%	244/421	[09:34<06:49, 2.31s/it]
Batches: 58%	245/421	[09:36<06:43, 2.29s/it]
Batches: 58%	246/421	[09:39<06:58, 2.39s/it]
Batches: 59%	247/421	[09:42<07:05, 2.45s/it]
Batches: 59%	248/421	[09:44<07:04, 2.45s/it]

Batches: 59%		249/421	[09:47<07:08,	2.49s/it]
Batches: 59%		250/421	[09:49<06:56,	2.44s/it]
Batches: 60%		251/421	[09:51<06:50,	2.41s/it]
Batches: 60%		252/421	[09:53<06:37,	2.35s/it]
Batches: 60%		253/421	[09:56<06:40,	2.38s/it]
Batches: 60%		254/421	[09:58<06:36,	2.38s/it]
Batches: 61%		255/421	[10:01<06:34,	2.38s/it]
Batches: 61%		256/421	[10:03<06:32,	2.38s/it]
Batches: 61%		257/421	[10:05<06:28,	2.37s/it]
Batches: 61%		258/421	[10:08<06:30,	2.40s/it]
Batches: 62%		259/421	[10:10<06:27,	2.39s/it]
Batches: 62%		260/421	[10:13<06:23,	2.38s/it]
Batches: 62%		261/421	[10:15<06:18,	2.36s/it]
Batches: 62%		262/421	[10:17<05:58,	2.26s/it]
Batches: 62%		263/421	[10:19<05:58,	2.27s/it]
Batches: 63%		264/421	[10:22<06:01,	2.30s/it]
Batches: 63%		265/421	[10:24<05:59,	2.30s/it]
Batches: 63%		266/421	[10:26<05:56,	2.30s/it]
Batches: 63%		267/421	[10:29<05:57,	2.32s/it]
Batches: 64%		268/421	[10:31<05:48,	2.27s/it]
Batches: 64%		269/421	[10:33<05:51,	2.31s/it]
Batches: 64%		270/421	[10:35<05:43,	2.27s/it]
Batches: 64%		271/421	[10:38<05:54,	2.36s/it]
Batches: 65%		272/421	[10:40<05:51,	2.36s/it]
Batches: 65%		273/421	[10:43<05:54,	2.40s/it]
Batches: 65%		274/421	[10:45<05:50,	2.38s/it]
Batches: 65%		275/421	[10:47<05:43,	2.35s/it]
Batches: 66%		276/421	[10:50<05:43,	2.37s/it]
Batches: 66%		277/421	[10:52<05:48,	2.42s/it]
Batches: 66%		278/421	[10:55<05:39,	2.38s/it]
Batches: 66%		279/421	[10:57<05:40,	2.40s/it]
Batches: 67%		280/421	[11:00<05:45,	2.45s/it]
Batches: 67%		281/421	[11:02<05:40,	2.43s/it]
Batches: 67%		282/421	[11:04<05:38,	2.44s/it]
Batches: 67%		283/421	[11:07<05:36,	2.44s/it]
Batches: 67%		284/421	[11:09<05:27,	2.39s/it]
Batches: 68%		285/421	[11:11<05:20,	2.36s/it]
Batches: 68%		286/421	[11:14<05:13,	2.32s/it]
Batches: 68%		287/421	[11:16<05:13,	2.34s/it]
Batches: 68%		288/421	[11:18<05:15,	2.37s/it]
Batches: 69%		289/421	[11:21<05:05,	2.32s/it]
Batches: 69%		290/421	[11:23<05:04,	2.32s/it]
Batches: 69%		291/421	[11:25<04:56,	2.28s/it]
Batches: 69%		292/421	[11:27<04:54,	2.28s/it]
Batches: 70%		293/421	[11:30<04:53,	2.29s/it]
Batches: 70%		294/421	[11:32<04:54,	2.32s/it]
Batches: 70%		295/421	[11:35<04:54,	2.34s/it]
Batches: 70%		296/421	[11:37<04:48,	2.31s/it]
Batches: 71%		297/421	[11:39<04:50,	2.34s/it]
Batches: 71%		298/421	[11:41<04:44,	2.32s/it]
Batches: 71%		299/421	[11:44<04:44,	2.34s/it]
Batches: 71%		300/421	[11:46<04:33,	2.26s/it]
Batches: 71%		301/421	[11:49<04:43,	2.36s/it]
Batches: 72%		302/421	[11:51<04:42,	2.38s/it]
Batches: 72%		303/421	[11:53<04:42,	2.39s/it]
Batches: 72%		304/421	[11:56<04:36,	2.36s/it]
Batches: 72%		305/421	[11:58<04:32,	2.35s/it]
Batches: 73%		306/421	[12:00<04:24,	2.30s/it]
Batches: 73%		307/421	[12:03<04:26,	2.34s/it]
Batches: 73%		308/421	[12:05<04:30,	2.39s/it]
Batches: 73%		309/421	[12:08<04:31,	2.42s/it]
Batches: 74%		310/421	[12:10<04:27,	2.41s/it]
Batches: 74%		311/421	[12:12<04:26,	2.43s/it]
Batches: 74%		312/421	[12:15<04:20,	2.39s/it]
Batches: 74%		313/421	[12:17<04:25,	2.46s/it]
Batches: 75%		314/421	[12:20<04:22,	2.46s/it]
Batches: 75%		315/421	[12:22<04:19,	2.45s/it]
Batches: 75%		316/421	[12:25<04:18,	2.47s/it]
Batches: 75%		317/421	[12:27<04:15,	2.46s/it]

Batches: 76%		318/421	[12:29<04:02,	2.35s/it]
Batches: 76%		319/421	[12:32<04:03,	2.39s/it]
Batches: 76%		320/421	[12:34<03:52,	2.30s/it]
Batches: 76%		321/421	[12:36<03:54,	2.35s/it]
Batches: 76%		322/421	[12:39<03:50,	2.32s/it]
Batches: 77%		323/421	[12:41<03:46,	2.31s/it]
Batches: 77%		324/421	[12:43<03:42,	2.30s/it]
Batches: 77%		325/421	[12:45<03:31,	2.20s/it]
Batches: 77%		326/421	[12:48<03:35,	2.26s/it]
Batches: 78%		327/421	[12:50<03:35,	2.29s/it]
Batches: 78%		328/421	[12:52<03:35,	2.32s/it]
Batches: 78%		329/421	[12:55<03:35,	2.34s/it]
Batches: 78%		330/421	[12:57<03:36,	2.38s/it]
Batches: 79%		331/421	[12:59<03:32,	2.36s/it]
Batches: 79%		332/421	[13:02<03:30,	2.37s/it]
Batches: 79%		333/421	[13:04<03:29,	2.38s/it]
Batches: 79%		334/421	[13:07<03:23,	2.34s/it]
Batches: 80%		335/421	[13:09<03:22,	2.35s/it]
Batches: 80%		336/421	[13:11<03:19,	2.35s/it]
Batches: 80%		337/421	[13:13<03:11,	2.27s/it]
Batches: 80%		338/421	[13:16<03:14,	2.35s/it]
Batches: 81%		339/421	[13:19<03:20,	2.45s/it]
Batches: 81%		340/421	[13:21<03:13,	2.39s/it]
Batches: 81%		341/421	[13:23<03:08,	2.36s/it]
Batches: 81%		342/421	[13:25<03:02,	2.31s/it]
Batches: 81%		343/421	[13:28<03:01,	2.33s/it]
Batches: 82%		344/421	[13:30<02:59,	2.33s/it]
Batches: 82%		345/421	[13:32<02:59,	2.36s/it]
Batches: 82%		346/421	[13:35<02:55,	2.34s/it]
Batches: 82%		347/421	[13:37<02:54,	2.36s/it]
Batches: 83%		348/421	[13:39<02:46,	2.28s/it]
Batches: 83%		349/421	[13:42<02:51,	2.38s/it]
Batches: 83%		350/421	[13:44<02:50,	2.41s/it]
Batches: 83%		351/421	[13:47<02:46,	2.37s/it]
Batches: 84%		352/421	[13:49<02:45,	2.41s/it]
Batches: 84%		353/421	[13:52<02:44,	2.42s/it]
Batches: 84%		354/421	[13:54<02:42,	2.42s/it]
Batches: 84%		355/421	[13:57<02:43,	2.47s/it]
Batches: 85%		356/421	[13:59<02:39,	2.45s/it]
Batches: 85%		357/421	[14:01<02:33,	2.40s/it]
Batches: 85%		358/421	[14:04<02:30,	2.39s/it]
Batches: 85%		359/421	[14:06<02:30,	2.42s/it]
Batches: 86%		360/421	[14:08<02:25,	2.38s/it]
Batches: 86%		361/421	[14:11<02:23,	2.39s/it]
Batches: 86%		362/421	[14:13<02:19,	2.36s/it]
Batches: 86%		363/421	[14:15<02:15,	2.33s/it]
Batches: 86%		364/421	[14:18<02:10,	2.29s/it]
Batches: 87%		365/421	[14:20<02:08,	2.30s/it]
Batches: 87%		366/421	[14:22<02:08,	2.34s/it]
Batches: 87%		367/421	[14:25<02:05,	2.32s/it]
Batches: 87%		368/421	[14:27<02:02,	2.31s/it]
Batches: 88%		369/421	[14:29<01:58,	2.29s/it]
Batches: 88%		370/421	[14:31<01:54,	2.24s/it]
Batches: 88%		371/421	[14:34<01:53,	2.26s/it]
Batches: 88%		372/421	[14:36<01:48,	2.21s/it]
Batches: 89%		373/421	[14:38<01:46,	2.23s/it]
Batches: 89%		374/421	[14:40<01:46,	2.26s/it]
Batches: 89%		375/421	[14:43<01:44,	2.28s/it]
Batches: 89%		376/421	[14:45<01:45,	2.33s/it]
Batches: 90%		377/421	[14:47<01:43,	2.36s/it]
Batches: 90%		378/421	[14:50<01:41,	2.35s/it]
Batches: 90%		379/421	[14:52<01:38,	2.34s/it]
Batches: 90%		380/421	[14:54<01:35,	2.32s/it]
Batches: 90%		381/421	[14:57<01:32,	2.32s/it]
Batches: 91%		382/421	[14:59<01:33,	2.41s/it]
Batches: 91%		383/421	[15:02<01:30,	2.39s/it]
Batches: 91%		384/421	[15:04<01:27,	2.36s/it]
Batches: 91%		385/421	[15:06<01:26,	2.40s/it]
Batches: 92%		386/421	[15:09<01:22,	2.36s/it]

Batches: 92%		387/421	[15:11<01:21,	2.40s/it]
Batches: 92%		388/421	[15:13<01:18,	2.37s/it]
Batches: 92%		389/421	[15:16<01:14,	2.34s/it]
Batches: 93%		390/421	[15:18<01:13,	2.37s/it]
Batches: 93%		391/421	[15:20<01:10,	2.34s/it]
Batches: 93%		392/421	[15:23<01:08,	2.35s/it]
Batches: 93%		393/421	[15:25<01:05,	2.34s/it]
Batches: 94%		394/421	[15:28<01:04,	2.39s/it]
Batches: 94%		395/421	[15:30<01:00,	2.31s/it]
Batches: 94%		396/421	[15:32<00:58,	2.35s/it]
Batches: 94%		397/421	[15:35<00:57,	2.38s/it]
Batches: 95%		398/421	[15:37<00:55,	2.43s/it]
Batches: 95%		399/421	[15:40<00:52,	2.40s/it]
Batches: 95%		400/421	[15:42<00:49,	2.38s/it]
Batches: 95%		401/421	[15:44<00:46,	2.31s/it]
Batches: 95%		402/421	[15:46<00:42,	2.26s/it]
Batches: 96%		403/421	[15:49<00:41,	2.29s/it]
Batches: 96%		404/421	[15:51<00:40,	2.36s/it]
Batches: 96%		405/421	[15:53<00:37,	2.33s/it]
Batches: 96%		406/421	[15:56<00:34,	2.33s/it]
Batches: 97%		407/421	[15:58<00:31,	2.28s/it]
Batches: 97%		408/421	[16:00<00:30,	2.33s/it]
Batches: 97%		409/421	[16:03<00:27,	2.33s/it]
Batches: 97%		410/421	[16:05<00:25,	2.29s/it]
Batches: 98%		411/421	[16:07<00:23,	2.30s/it]
Batches: 98%		412/421	[16:10<00:21,	2.42s/it]
Batches: 98%		413/421	[16:12<00:18,	2.35s/it]
Batches: 98%		414/421	[16:14<00:16,	2.33s/it]
Batches: 99%		415/421	[16:17<00:14,	2.36s/it]
Batches: 99%		416/421	[16:19<00:11,	2.35s/it]
Batches: 99%		417/421	[16:22<00:09,	2.42s/it]
Batches: 99%		418/421	[16:24<00:07,	2.34s/it]
Batches: 100%		419/421	[16:26<00:04,	2.31s/it]
Batches: 100%		420/421	[16:28<00:02,	2.33s/it]
Batches: 100%		421/421	[16:30<00:00,	2.26s/it]
Batches: 100%		421/421	[16:30<00:00,	2.35s/it]

=====

✓ Successfully processed: 3366/3366
x Skipped (missing/error): 0/3366

Extracted shapes:
Positive: (3366, 3584)
Negative: (3366, 3584)
Labels: (3366,)

Cached to:
hidden_states_cache_final/cache_attribute_recognition_3366_qwen2_ccs_aligned.npz

Logistic regression sanity-check accuracy: 77.8%

=====

TRAINING CCS PROBE

=====

Dataset split:
Train: 2019 samples (1030 pos, 989 neg)
Test: 1347 samples (653 pos, 694 neg)
Hidden dim: 3584

=====

TRAINING WITH MULTIPLE RANDOM RESTARTS

=====

Trial 1/10: Loss = 0.007007
Trial 2/10: Loss = 0.002527
Trial 3/10: Loss = 0.001526
Trial 4/10: Loss = 0.003509
Trial 5/10: Loss = 0.011409
Trial 6/10: Loss = 0.001516

Trial 7/10: Loss = 0.003003
Trial 8/10: Loss = 0.004485
Trial 9/10: Loss = 0.002015
Trial 10/10: Loss = 0.003503

=====

EVALUATION WITH BEST PROBE

=====

Best loss: 0.001516

Test Results:

Overall Accuracy: 82.0% (1104/1347)
Positive samples: 84.5% (653 samples)
Negative samples: 79.5% (694 samples)

✓ COMPLETE: attribute_recognition → 82.0%

CATEGORY: SPATIAL_RECOGNITION
#####

=====

EXTRACTING HIDDEN STATES: SPATIAL_RECOGNITION

=====

⚠ No cache found. Starting extraction...

Processing 976 samples in batches of 8
Searching in 2 image directories
LOADING MODEL: qwen2
Device: cuda

Fetching 5 files: 0%|██████████| 0/5 [00:00<?, ?it/s]
Fetching 5 files: 100%|██████████| 5/5 [00:00<00:00, 1398.57it/s]

Loading weights: 0%|██████████| 0/730 [00:00<?, ?it/s]
Loading weights: 0%|██████████| 1/730 [00:00<05:36, 2.17it/s]
Loading weights: 0%|██████████| 2/730 [00:01<06:09, 1.97it/s]
Loading weights: 1%|██████████| 5/730 [00:01<02:07, 5.67it/s]
Loading weights: 2%|██████████| 16/730 [00:01<00:38, 18.71it/s]
Loading weights: 4%|██████████| 28/730 [00:01<00:24, 29.09it/s]
Loading weights: 5%|██████████| 40/730 [00:01<00:18, 36.39it/s]
Loading weights: 7%|██████████| 52/730 [00:01<00:14, 45.48it/s]
Loading weights: 8%|██████████| 58/730 [00:02<00:15, 44.52it/s]
Loading weights: 9%|██████████| 65/730 [00:02<00:14, 45.38it/s]
Loading weights: 10%|██████████| 76/730 [00:02<00:14, 46.60it/s]
Loading weights: 12%|██████████| 88/730 [00:02<00:13, 48.80it/s]
Loading weights: 14%|██████████| 100/730 [00:02<00:12, 50.15it/s]
Loading weights: 15%|██████████| 112/730 [00:03<00:12, 51.08it/s]
Loading weights: 17%|██████████| 124/730 [00:03<00:10, 56.91it/s]
Loading weights: 18%|██████████| 130/730 [00:03<00:11, 53.22it/s]
Loading weights: 19%|██████████| 137/730 [00:03<00:11, 51.89it/s]
Loading weights: 20%|██████████| 148/730 [00:03<00:10, 56.68it/s]
Loading weights: 21%|██████████| 154/730 [00:03<00:10, 52.38it/s]
Loading weights: 22%|██████████| 161/730 [00:04<00:11, 51.15it/s]
Loading weights: 24%|██████████| 172/730 [00:04<00:11, 48.78it/s]
Loading weights: 25%|██████████| 180/730 [00:04<00:10, 54.69it/s]
Loading weights: 25%|██████████| 186/730 [00:04<00:11, 46.25it/s]
Loading weights: 27%|██████████| 197/730 [00:04<00:10, 52.96it/s]
Loading weights: 28%|██████████| 207/730 [00:04<00:09, 56.12it/s]
Loading weights: 29%|██████████| 213/730 [00:05<00:09, 53.37it/s]
Loading weights: 30%|██████████| 221/730 [00:05<00:09, 51.21it/s]
Loading weights: 32%|██████████| 230/730 [00:05<00:08, 59.24it/s]
Loading weights: 32%|██████████| 237/730 [00:05<00:09, 50.56it/s]
Loading weights: 34%|██████████| 245/730 [00:05<00:09, 50.92it/s]
Loading weights: 35%|██████████| 256/730 [00:05<00:09, 50.57it/s]
Loading weights: 37%|██████████| 268/730 [00:06<00:08, 56.89it/s]
Loading weights: 38%|██████████| 274/730 [00:06<00:08, 51.33it/s]
Loading weights: 38%|██████████| 281/730 [00:06<00:09, 45.42it/s]
Loading weights: 40%|██████████| 293/730 [00:06<00:08, 53.99it/s]

Loading weights: 42%		304/730	[00:06<00:07, 58.56it/s]
Loading weights: 43%		311/730	[00:06<00:07, 56.10it/s]
Loading weights: 43%		317/730	[00:07<00:08, 50.47it/s]
Loading weights: 45%		325/730	[00:07<00:07, 56.70it/s]
Loading weights: 45%		331/730	[00:07<00:08, 47.15it/s]
Loading weights: 52%		383/730	[00:07<00:02, 147.15it/s]
Loading weights: 62%		450/730	[00:07<00:01, 267.41it/s]
Loading weights: 70%		511/730	[00:07<00:00, 351.01it/s]
Loading weights: 79%		575/730	[00:07<00:00, 425.89it/s]
Loading weights: 88%		643/730	[00:07<00:00, 492.17it/s]
Loading weights: 97%		705/730	[00:07<00:00, 527.48it/s]
Loading weights: 100%		730/730	[00:08<00:00, 91.02it/s]

✓ Model loaded successfully

Batches: 0%		0/122	[00:00<?, ?it/s]
Batches: 1%		1/122	[00:02<04:19, 2.15s/it]
Batches: 2%		2/122	[00:04<04:18, 2.16s/it]
Batches: 2%		3/122	[00:06<04:38, 2.34s/it]
Batches: 3%		4/122	[00:09<04:37, 2.35s/it]
Batches: 4%		5/122	[00:11<04:35, 2.36s/it]
Batches: 5%		6/122	[00:13<04:30, 2.33s/it]
Batches: 6%		7/122	[00:16<04:25, 2.31s/it]
Batches: 7%		8/122	[00:18<04:24, 2.32s/it]
Batches: 7%		9/122	[00:20<04:24, 2.34s/it]
Batches: 8%		10/122	[00:23<04:24, 2.36s/it]
Batches: 9%		11/122	[00:25<04:27, 2.41s/it]
Batches: 10%		12/122	[00:28<04:22, 2.38s/it]
Batches: 11%		13/122	[00:30<04:17, 2.37s/it]
Batches: 11%		14/122	[00:32<04:11, 2.33s/it]
Batches: 12%		15/122	[00:35<04:16, 2.40s/it]
Batches: 13%		16/122	[00:37<04:16, 2.42s/it]
Batches: 14%		17/122	[00:40<04:15, 2.44s/it]
Batches: 15%		18/122	[00:42<04:12, 2.43s/it]
Batches: 16%		19/122	[00:44<04:01, 2.34s/it]
Batches: 16%		20/122	[00:47<04:05, 2.41s/it]
Batches: 17%		21/122	[00:49<03:55, 2.33s/it]
Batches: 18%		22/122	[00:52<04:00, 2.41s/it]
Batches: 19%		23/122	[00:54<04:00, 2.43s/it]
Batches: 20%		24/122	[00:56<03:52, 2.37s/it]
Batches: 20%		25/122	[00:59<03:51, 2.38s/it]
Batches: 21%		26/122	[01:01<03:58, 2.48s/it]
Batches: 22%		27/122	[01:04<03:52, 2.45s/it]
Batches: 23%		28/122	[01:06<03:42, 2.36s/it]
Batches: 24%		29/122	[01:08<03:36, 2.32s/it]
Batches: 25%		30/122	[01:10<03:33, 2.32s/it]
Batches: 25%		31/122	[01:13<03:29, 2.30s/it]
Batches: 26%		32/122	[01:15<03:30, 2.34s/it]
Batches: 27%		33/122	[01:17<03:26, 2.32s/it]
Batches: 28%		34/122	[01:20<03:31, 2.40s/it]
Batches: 29%		35/122	[01:22<03:27, 2.38s/it]
Batches: 30%		36/122	[01:25<03:23, 2.37s/it]
Batches: 30%		37/122	[01:27<03:18, 2.34s/it]
Batches: 31%		38/122	[01:29<03:15, 2.33s/it]
Batches: 32%		39/122	[01:32<03:14, 2.34s/it]
Batches: 33%		40/122	[01:34<03:11, 2.34s/it]
Batches: 34%		41/122	[01:36<03:08, 2.33s/it]
Batches: 34%		42/122	[01:39<03:04, 2.30s/it]
Batches: 35%		43/122	[01:41<02:57, 2.24s/it]
Batches: 36%		44/122	[01:43<02:56, 2.27s/it]
Batches: 37%		45/122	[01:45<02:53, 2.26s/it]
Batches: 38%		46/122	[01:47<02:48, 2.22s/it]
Batches: 39%		47/122	[01:50<02:48, 2.25s/it]
Batches: 39%		48/122	[01:52<02:46, 2.26s/it]
Batches: 40%		49/122	[01:54<02:51, 2.35s/it]
Batches: 41%		50/122	[01:57<02:47, 2.33s/it]
Batches: 42%		51/122	[01:59<02:47, 2.36s/it]
Batches: 43%		52/122	[02:01<02:42, 2.33s/it]
Batches: 43%		53/122	[02:04<02:39, 2.31s/it]

Batches: 44%	54/122	[02:06<02:34,	2.27s/it]
Batches: 45%	55/122	[02:08<02:34,	2.30s/it]
Batches: 46%	56/122	[02:11<02:31,	2.30s/it]
Batches: 47%	57/122	[02:13<02:28,	2.28s/it]
Batches: 48%	58/122	[02:15<02:29,	2.34s/it]
Batches: 48%	59/122	[02:18<02:28,	2.36s/it]
Batches: 49%	60/122	[02:20<02:27,	2.38s/it]
Batches: 50%	61/122	[02:22<02:24,	2.37s/it]
Batches: 51%	62/122	[02:25<02:25,	2.43s/it]
Batches: 52%	63/122	[02:27<02:21,	2.40s/it]
Batches: 52%	64/122	[02:30<02:17,	2.37s/it]
Batches: 53%	65/122	[02:32<02:14,	2.37s/it]
Batches: 54%	66/122	[02:34<02:10,	2.34s/it]
Batches: 55%	67/122	[02:37<02:09,	2.35s/it]
Batches: 56%	68/122	[02:39<02:03,	2.30s/it]
Batches: 57%	69/122	[02:41<02:03,	2.33s/it]
Batches: 57%	70/122	[02:44<02:01,	2.33s/it]
Batches: 58%	71/122	[02:46<02:00,	2.35s/it]
Batches: 59%	72/122	[02:48<01:58,	2.38s/it]
Batches: 60%	73/122	[02:51<01:54,	2.33s/it]
Batches: 61%	74/122	[02:53<01:52,	2.34s/it]
Batches: 61%	75/122	[02:55<01:51,	2.37s/it]
Batches: 62%	76/122	[02:58<01:48,	2.37s/it]
Batches: 63%	77/122	[03:00<01:44,	2.32s/it]
Batches: 64%	78/122	[03:02<01:42,	2.33s/it]
Batches: 65%	79/122	[03:05<01:41,	2.35s/it]
Batches: 66%	80/122	[03:07<01:39,	2.36s/it]
Batches: 66%	81/122	[03:10<01:39,	2.43s/it]
Batches: 67%	82/122	[03:12<01:37,	2.43s/it]
Batches: 68%	83/122	[03:15<01:33,	2.40s/it]
Batches: 69%	84/122	[03:17<01:31,	2.42s/it]
Batches: 70%	85/122	[03:20<01:31,	2.47s/it]
Batches: 70%	86/122	[03:22<01:26,	2.42s/it]
Batches: 71%	87/122	[03:24<01:21,	2.34s/it]
Batches: 72%	88/122	[03:26<01:20,	2.37s/it]
Batches: 73%	89/122	[03:29<01:18,	2.37s/it]
Batches: 74%	90/122	[03:31<01:16,	2.38s/it]
Batches: 75%	91/122	[03:34<01:13,	2.36s/it]
Batches: 75%	92/122	[03:36<01:11,	2.39s/it]
Batches: 76%	93/122	[03:38<01:08,	2.35s/it]
Batches: 77%	94/122	[03:41<01:09,	2.47s/it]
Batches: 78%	95/122	[03:43<01:04,	2.40s/it]
Batches: 79%	96/122	[03:46<01:01,	2.37s/it]
Batches: 80%	97/122	[03:48<00:58,	2.34s/it]
Batches: 80%	98/122	[03:50<00:56,	2.34s/it]
Batches: 81%	99/122	[03:53<00:53,	2.35s/it]
Batches: 82%	100/122	[03:55<00:50,	2.30s/it]
Batches: 83%	101/122	[03:57<00:49,	2.35s/it]
Batches: 84%	102/122	[03:59<00:46,	2.32s/it]
Batches: 84%	103/122	[04:02<00:43,	2.31s/it]
Batches: 85%	104/122	[04:04<00:41,	2.30s/it]
Batches: 86%	105/122	[04:06<00:39,	2.30s/it]
Batches: 87%	106/122	[04:09<00:37,	2.31s/it]
Batches: 88%	107/122	[04:11<00:34,	2.31s/it]
Batches: 89%	108/122	[04:13<00:32,	2.33s/it]
Batches: 89%	109/122	[04:16<00:30,	2.34s/it]
Batches: 90%	110/122	[04:18<00:27,	2.28s/it]
Batches: 91%	111/122	[04:20<00:25,	2.31s/it]
Batches: 92%	112/122	[04:22<00:22,	2.29s/it]
Batches: 93%	113/122	[04:25<00:20,	2.31s/it]
Batches: 93%	114/122	[04:27<00:18,	2.36s/it]
Batches: 94%	115/122	[04:29<00:16,	2.30s/it]
Batches: 95%	116/122	[04:32<00:13,	2.31s/it]
Batches: 96%	117/122	[04:34<00:11,	2.24s/it]
Batches: 97%	118/122	[04:36<00:09,	2.30s/it]
Batches: 98%	119/122	[04:39<00:06,	2.28s/it]
Batches: 98%	120/122	[04:41<00:04,	2.32s/it]
Batches: 99%	121/122	[04:43<00:02,	2.35s/it]
Batches: 100%	122/122	[04:46<00:00,	2.43s/it]

Batches: 100%|██████████| 122/122 [04:46<00:00, 2.35s/it]

```
=====
✓ Successfully processed: 976/976
x Skipped (missing/error): 0/976
```

Extracted shapes:
Positive: (976, 3584)
Negative: (976, 3584)
Labels: (976,)

Cached to: hidden_states_cache_final/cache_spatial_recognition_976_qwen2_ccs_aligned.npz

Logistic regression sanity-check accuracy: 78.3%

```
=====
TRAINING CCS PROBE
=====
```

Dataset split:
Train: 585 samples (289 pos, 296 neg)
Test: 391 samples (199 pos, 192 neg)
Hidden dim: 3584

```
=====
TRAINING WITH MULTIPLE RANDOM RESTARTS
=====
```

```
Trial 1/10: Loss = 0.010261
Trial 2/10: Loss = 0.005145
Trial 3/10: Loss = 0.013680
Trial 4/10: Loss = 0.005135
Trial 5/10: Loss = 0.015390
Trial 6/10: Loss = 0.010237
Trial 7/10: Loss = 0.008552
Trial 8/10: Loss = 0.005134
Trial 9/10: Loss = 0.017101
Trial 10/10: Loss = 0.003426
```

```
=====
EVALUATION WITH BEST PROBE
=====
```

Best loss: 0.003426

Test Results:
Overall Accuracy: 74.7% (292/391)
Positive samples: 73.9% (199 samples)
Negative samples: 75.5% (192 samples)

✓ COMPLETE: spatial_recognition → 74.7%

```
=====
Final Results:
object_detection      : 89.9%
attribute_recognition : 82.0%
spatial_recognition   : 74.7%

Average               : 82.2%
=====
```

=== linear_ccs.py finished at Sat Sep 19 10:46:55 CEST 2026 with exit code: 0 ===